

Name: _____ Partner: _____ Period: _____

Missing Data Points

A scientist's measurement device records four data streams, each modeled by a rational function. At certain input values, the device has a gap in the data. Your job is to analyze each function, determine whether each gap is a **hole** or a **vertical asymptote**, and — if it is a hole — find the exact coordinates.

The four functions are:

$$f_1(x) = \frac{x^2 - 9}{x - 3} \quad f_2(x) = \frac{x^2 - 1}{(x - 1)(x + 4)} \quad f_3(x) = \frac{(x - 2)^2}{x^2 - 4} \quad f_4(x) = \frac{x^2 + 2x}{x}$$

Tier R — Remediate

The numerator and denominator of each function are already written in a helpful form below. For each function, circle the shared factor, then decide: is $x = c$ a **hole** or a **VA**?

Fn	Numerator (factored)	Denominator (factored)	Gap at	Hole or VA?
f_1	$(x - 3)(x + 3)$	$(x - 3)$	$x = 3$	_____
f_2	$(x - 1)(x + 1)$	$(x - 1)(x + 4)$	$x = 1$	_____
			$x = -4$	_____
f_3	$(x - 2)^2$	$(x - 2)(x + 2)$	$x = 2$	_____
			$x = -2$	_____
f_4	$x(x + 2)$	x	$x = 0$	_____

Tier A — Apply

- Factor the numerator and denominator of each function yourself. Show your work. Then identify all shared zeros and apply the multiplicity rule to classify each as a hole or VA.
- For each hole you identified, find the coordinates (c, L) . Show the cancellation and substitution steps.

Function	Shared zero c	Simplified form	Hole at (c, L)
f_1	_____	_____	_____
f_2	_____	_____	_____
f_3	_____	_____	_____
f_4	_____	_____	_____

Tier E — Extend

1. Write the limit notation for each hole from Tier A. For example: $\lim_{x \rightarrow c} f_i(x) = L$.

$$\lim_{x \rightarrow \underline{\quad}} f_1(x) = \underline{\quad} \quad \lim_{x \rightarrow \underline{\quad}} f_2(x) = \underline{\quad} \quad \lim_{x \rightarrow \underline{\quad}} f_3(x) = \underline{\quad}$$

$$\lim_{x \rightarrow \underline{\quad}} f_4(x) = \underline{\quad}$$

2. Choose one hole from above. Explain in a complete sentence *why* the y -coordinate of the hole is that specific value, referencing what you did to find it.

3. Construct a rational function $g(x)$ that has **both** a hole at $x = 1$ **and** a vertical asymptote at $x = -2$. Write your answer in factored form and in simplified form.

Factored: $g(x) = \underline{\hspace{4cm}}$

Simplified (hole removed): $g(x) = \underline{\hspace{4cm}}$ with hole at $(\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$